

NASA Data Science & Astronomy Research Internship

Summer 2026

Exoplanet data analysis • NASA missions • Scientific writing

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Assignments to date

Older:

Telescope Observing
Review NASA mission study proposals, study reports, and mission proposals
Run the AU Mic jupyter notebook; Experiment with the code
Pick a TOI from the Exoplanet Archive
Retrieve a new TESS candidate light curve from MAST and generate a plot
Change time and flux ranges of the plot; Post plot to Discord Channel

Recent:

Download GMU telescope TOI follow-up data sets, analyze and produce light curves.
Post progress and challenges to discord
Research your TOI on the NASA Exoplanet Archive
Prepare presentation poster for symposium

Future:

Write your paper

Optional:

Pick a JWST transit observation (spreadsheet of options forthcoming)
Analyze following the Jupyter notebook tutorial
Write a separate paper on this observation (examples forthcoming)

Access to all data:

Go to spreadsheet shared with you via email and Discord. Please enter your name in Column N, first and last initial. This is NOT first come, first serve, so don't rush. Multiple people can analyze the same data set, together as a team of up to 3, or independently. Put a ";" (semi-colon) between names. If a group, indicate this as well with some explanation.

To assist in picking a data set, look at it first. research the TOI on the NASA Exoplanet archive. google the target. Ask AI. Pay particular attention to some of our team's old notes in column S.

2022 data: <https://drive.google.com/drive/folders/148QIVQoSZVHBrvJmaSV5IGlpspOYdsGK?usp=sharing>

2023 data: https://drive.google.com/drive/folders/1baHyEpGMZzcPr6OnAspDXhS9Ho_67IUq?usp=sharing

2024 data: <https://drive.google.com/drive/folders/1HZyoRaUZarzPpO8kOLwQkd90M7mpeHRU?usp=sharing>

2025 data: https://drive.google.com/drive/folders/1uI4uYjjzxeS5g7ewToQfrW48p_34Z3HA?usp=sharing

Use the date from the above spreadsheet, and go into that folder. You should find a subfolder with the same name as the TOI number. Download that folder and its data. Extract the multiple 2GB zip files, and merge the file contents.

What about JWST data

- Use the MAST archive to retrieve
- Attempt only after receiving approval for your light curve analysis via Discord for the campus telescope TESS follow-up data.
- Why? JWST data is another level deeper in complexity.
- What the video sent previously for how to access the data, how to access code to process and analyze JWST data, ... or ... write your own! It is better to be a tool creator than solely a tool user.
- Use the python jupyter notebook tutorials available on the STScI github account.

Schedule

June ~~13, 20, 27~~, July 11 – live lectures

~~June 25, July 1, July 4~~, 15 – asynchronous lectures

~~July 11, 18~~, 25 – live office hours

July 31st – Symposium tours

August 1st – Symposium day

Topics for Our Program

~~Introductions~~

~~Logistics~~

~~How a NASA Mission Happens~~

~~Working with data, part 1: Jupyter notebooks
and FITS files~~

~~Telescopes on the Ground and in Space
(part 2)~~

~~The NASA TESS mission and Transiting
Exoplanets~~

~~Working with data, part 2: AstrolImageJ~~

**Making a Measurement: Science Methods
and Inference**

~~Science Writing~~

Space Revolution and Space Policy

~~Working with data, part 3: JWST~~

Bonus? Mission Engineering

Draft Symposium Schedule

- 9-9:30: Checkin, set up posters (digital and print)
- 9:30-11:30: Talks
- 11:30-1:00: Poster viewing & lunch
- 1:00-3:00 – Research Lab Tours

Final homework Assignments: the poster

- Poster – 48x36” max – 40x30” template:
https://www.dropbox.com/scl/fi/awofmzrkfdj083r16oq0w/ScharScholars_NAME_PosterTemplate.pptx?rlkey=ljfyftt7r15vm206hewolegoq&st=y2uxh07s&dl=1 (password: Schar2026!)
- We will send out a survey to learn your preference for a poster or a talk; and talk selection depends on analysis posted to discord.
- How to make a prize winning poster:
<https://sites.psu.edu/astrowright/2013/09/17/make-award-winning-posters/>

Final homework Assignments: the talk

- PPT or Google Slides (keynote ok, but not preferred)
- 1 minute per slide, at most 2 slides per minute
- Practice your talks in advance → helps with flow, comfort level and timing (5 minutes max)
- Try to avoid large blocks of text. → put this in notes or a script
- Focus figures (including images and screen grabs), tables or bulleted lists
- Slides should follow the sections of the paper, with at most 2-3 slides per section, ideally 1-2.
 - Intro, Observations, Data Reduction, Aperture Photometry, Light Curve Detrending, Result, Discussion, Conclusions and Future work, Acknowledgments

Final homework Assignments: the paper

- Due August 29th
- Discord will remain open for Q&A, check-ins, assistance, technical support; I will not be responsive to email. I will not have time to review writing in general.
- Lean into each other on Discord:
 - Swap drafts!
 - Compare data reduction approaches;
 - Use the chat and audio channels on Discord
 - Get a study group together! Self-organize by high school or geographic region. You can achieve more together than working in isolation.
 - Post light curves to discord, and light curve settings, post xls target light curve files, and .plotcfg files to reproduce a plot.
 - My team will assist you as much as we can through when our semester starts (August 25th)

Final homework Assignments: the paper

Please send as a PDF to execed@gmu.edu and nasa.schar.program@gmail.com
Use Google Docs, Word, or Overleaf if you go the latex route; you can send these as well.

Your papers may be lightly edited and will be posted as a student journal from George Mason University. We will provide this link to you in the fall to use in your resumes and college applications.

Final homework Assignments: the paper

Example papers from last year: <https://science.gmu.edu/academics/departments-units/physics-and-astronomy-department/observatory/schar-astro-scholars>

Here are some professional examples on the validation of a transiting exoplanet with ground-based follow-up:

This is a good example from a high school student:

<https://emerginginvestigators.org/articles/ground-based-follow-up-observations-of-tess-exoplanet-candidates/pdf>

<https://arxiv.org/abs/2110.14344>

<https://academic.oup.com/mnras/article/516/3/4432/6692879>

<https://arxiv.org/abs/2210.08179>

<https://arxiv.org/abs/2205.05709>

<https://arxiv.org/abs/2212.08242>

<https://arxiv.org/abs/2208.07328>

This links to even more TESS follow-up papers I have been on:

<https://ui.adsabs.harvard.edu/search/q=%20author%3A%22Plavchan%22%20title%3A%22TOI%22&sort=date%20desc%2C%20bibcode%20desc&p=0>

Note: **DO NOT pay for any paper access.** If you want access to a particular paper, often George Mason has institutional access. Send me a link and I can see if I can provide you a free copy. When looking for papers via NASA ADS, look for the free “arXiv” preprint link.

Example websites and papers for reference:

<https://tess.mit.edu/followup/>

<https://arxiv.org/abs/2103.12538>

https://asd.gsfc.nasa.gov/archive/tess/ground_based_followup.html

Final homework Assignments: the paper

Science goals of paper:

Verify that the transit occurs on the TESS object of interest (you need a finder chart to know which is your target - https://astro.swarthmore.edu/transits/finding_charts.cgi , at the time predicted, with the depth (brightness change) expected, and with the duration expected.

If not, it could be poor timing knowledge (the timing uncertainty matters), or a false-positive for a variety of reasons.

We provided the expected start and stop times of transit for you, but once you have light curves for us to look at, post them to discord.

We can look up more details about the transit, such as the timing uncertainty, and notes from the TESS team on what other observations have revealed. Only I and Kevin have access to this TESS team information to share. Only when you have a light curve can we move on to interpretation and provide you this information.

If there is no detectable transit, it could be that the signal was too small given the noise in our lightcurve to discern it.

It could also be that there is a nearby star that is an eclipsing binary, and AstrolmageJ has a tool called the nearby eclipsing binary (NEB) check that we can help you run and learn to interpret.

Final homework Assignments: the paper

- Most importantly, try your best. Get as far as you can in the process, and write up what you are able to complete. If you fall short of our goals, conclude with a future work section outlining the remaining work to be completed in the future (by you or someone else)
- This is a very challenging assignment, one that usually not all college undergraduate students can complete successfully as part of a course.
- You will struggle – you’ve never done this before.
- You have very little help, mentorship and guidance. There is no known “right answer” unlike in a math class. We don’t know what the outcome will be.
- Each night’s data outcome can be very different!
 - Many may be plagued by clouds, data gaps, and result in an inconclusive outcome.
 - Many may result in marginal or noisy detections
 - Some transits may be incomplete (partial)
 - Some will be false positives.
 - Only a few will result in clear, on target detections
 - Be careful with the language you use in your papers – try to be precise
- The outcome may be more work that will have to be deferred to the future.
- You may answer one question but leave or create several more unanswered.
- ***This... is research.***

Topics for Our Program

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Basic Statistics

Mean:

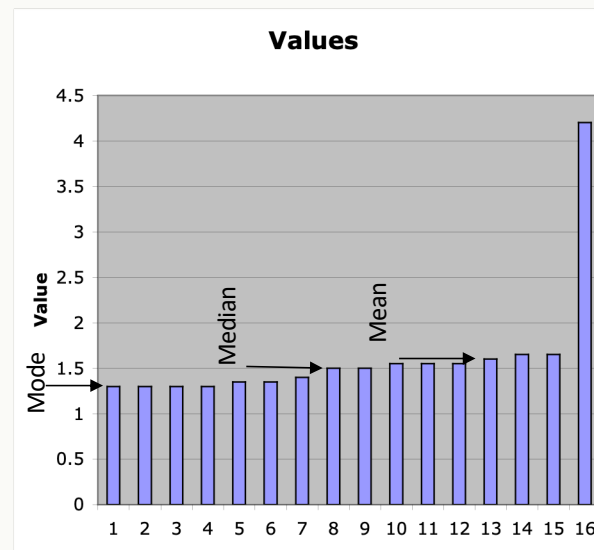
"Average value" = 1.63

$$\bar{x} = \frac{1}{N} \sum_i x_i$$

Median: *The individual value from the collection such that 1/2 the observations are less and 1/2 are greater = 1.5*

Median is often useful because it is less affected by a single outlier than the mean

Mode: *The most frequently occurring value in the collection = 1.3*



16 Values: 1.3, 1.3, 1.3, 1.3, 1.35, 1.35, 1.4, 1.5, 1.5, 1.55, 1.55, 1.55, 1.6, 1.65, 1.65, 4.2

How measurements are distributed

Most of what we know is derived by making measurements

A priori, when we make a set of measurements, we may know and expect what type of distribution of values of the measurements that we will get (more on this later).

However, we may not know the distributions “central value” (for example, mean), nor it’s size or scale.

Quantifying this distribution of measurements is known as quantifying uncertainty or “errors”.

The idea of error

It is never possible to measure anything exactly (eventually Heisenberg's uncertainty principle would bite you!).

So when you measure something, you have to say how precisely you have measured it -> This is the scale of the distribution of measurements.

Example:

Planck's constant h is known to approximately 1.5 parts in 10^8 , or $6.626070040(81) \times 10^{-34}$ J-s, e.g. $6.626070040 \pm 0.000000081 \times 10^{-34}$ J-s

Suppose you derive an experiment to measure Planck's constant. You can't just take their value and error and use as their own. The above is the work of other scientists' experiments. Your experiment will have its own value and associated error.

The idea of error

Errors are also not 'blunders' or experimental mishaps which can be fixed.

If a telescope is bumped during an exposure resulting in two sets of stars, the image is simply deleted and redone. If it's not caught at the time, then this datum is discarded.

Errors occur for all measurements. It is impossible to know exactly how far off a measurement is. Otherwise, the measurements would just be 'adjusted' (fudge-factor) to the proper value. But this is not possible nor good science.

The classification of error

Errors occur in two types: systematic and random.

Systematic errors are errors which shift all measurements in the same way. Things like improper equipment calibration, thicker atmosphere than anticipated, etc.

An example is a ruler with the wrong length.

The classification of error

Random errors displace measurements in an arbitrary direction (they scatter measurements about a value). They cannot be eliminated, but usually can be reduced (by making multiple measurements).

For example, each measurement of a star includes random fluctuations caused by seeing, detector noise, random sky noise, etc.

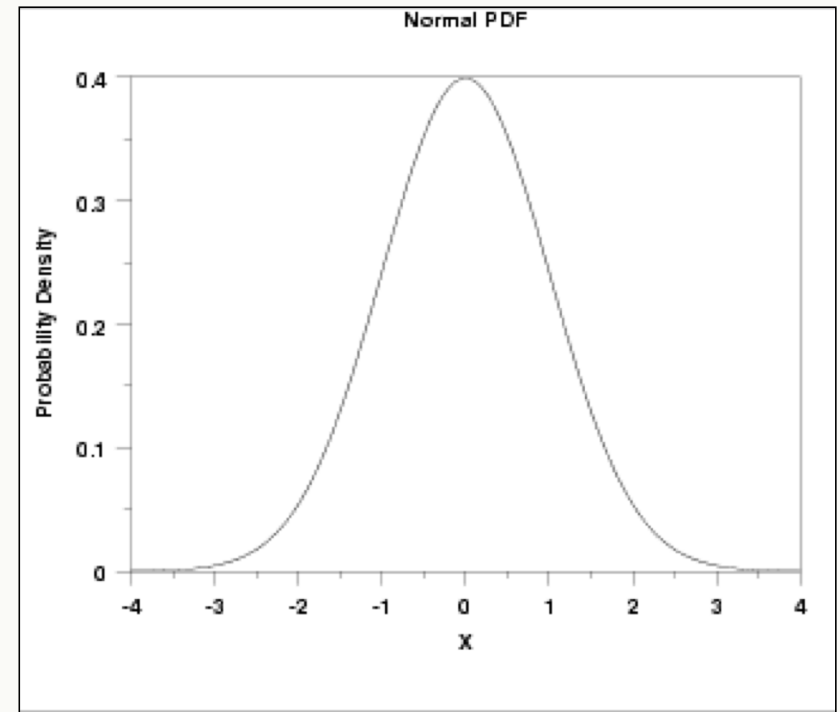
OK, so how do we quantify errors?

How sets of events or measurements are distributed

Often with data we have a large number of “events”. For most experiments, the outcome is a continuous distribution of possible values. In this case, the distribution of results approaches a smooth curve as the number of measurements approaches infinity.

$$y = a e^{[-(x-x_0)^2 / (2 \sigma^2)]}$$

Other continuous examples: uniform random distribution; lognormal distribution, student t-distribution, etc...



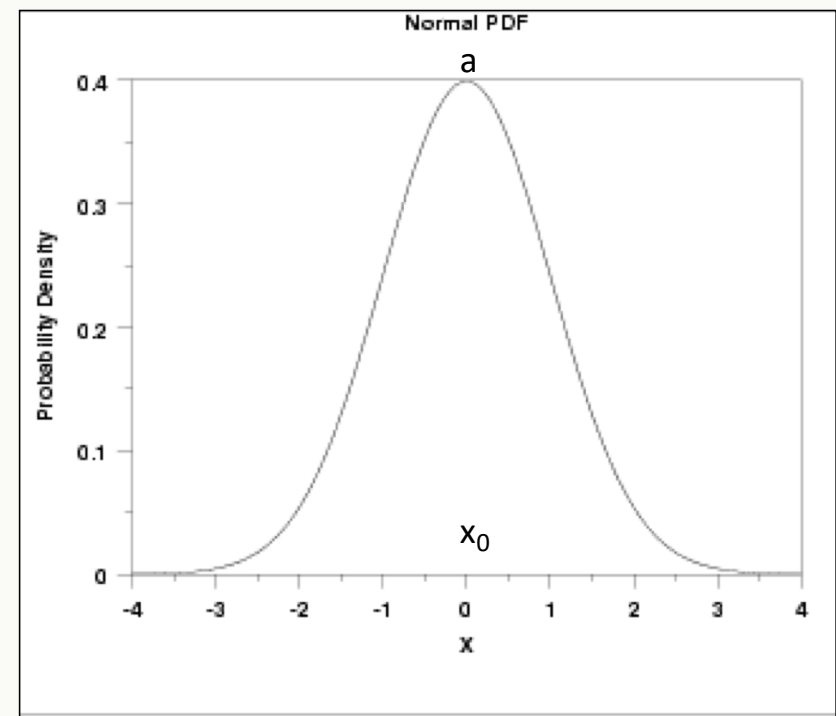
Gaussian Distribution: x_0 = mean, σ = standard deviation

Gaussian Distribution - FWHM

$$y = a e^{-(x-x_0)^2 / (2 \sigma^2)}$$

Full width at half maximum:

$$\begin{aligned} FWHM &= \sigma \cdot 2\sqrt{2\ln 2} \\ &= 2.35482 \cdot \sigma \end{aligned}$$



Binomial Distribution

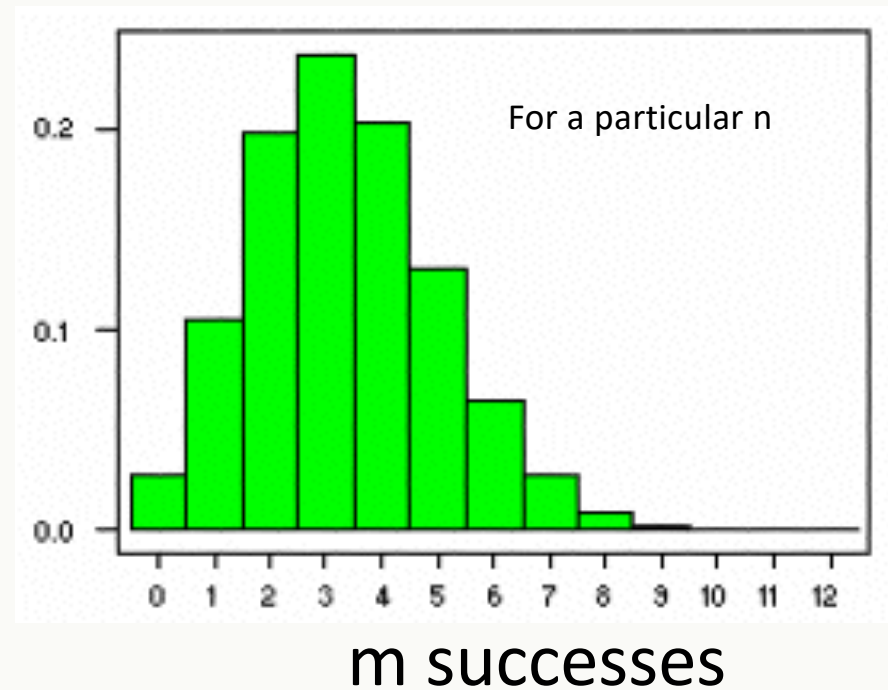
- Where there are exactly 2 mutually exclusive outcomes – success or failure.
- The probability of observing m successes in n observations

$$P(m, p, n) = \frac{n!}{m!(n-m)!} p^m (1-p)^{(n-m)}$$

- p is probability of success in a single observation

Approaches Gaussian for large n and m , but <0 excluded

p



Poisson Distribution – when the outcome is a discrete integer and positive definite

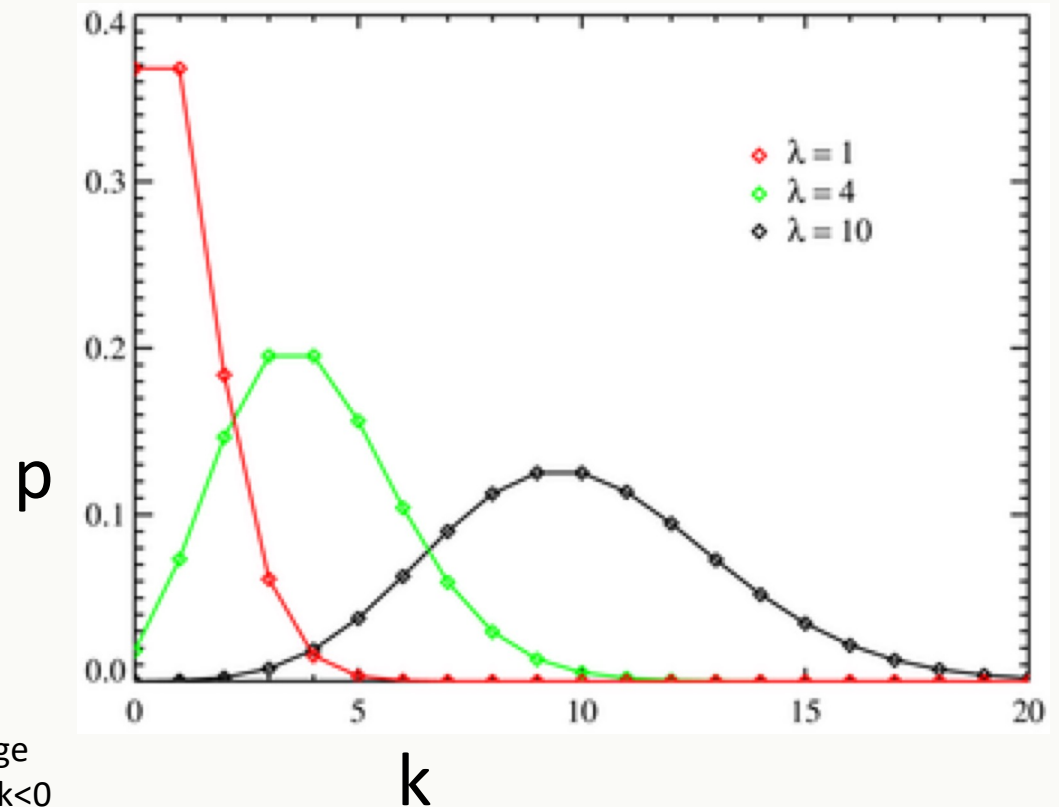
The limit of the binomial distribution for rare events and a large number of trials.

The “counting rate” distribution. The value can never be less than zero. For example the number of photons that arrive in a time interval can be 0, 1, 2, ... but can never be less than 0

Pr(k events given lambda) =

$$\frac{\lambda^k e^{-\lambda}}{k!}$$

Similar to Gaussian for large lambda, but still excludes $k < 0$ and only integer measurements



Estimating How Good the Data Are

(Simple) deviation: (Note that $\bar{x}$ is how we represent the mean)

$$d_i = x_i - \bar{x}$$

Mean deviation: The N-1 represents the loss of one degree of freedom - with only 1 data point the mean deviation is meaningless

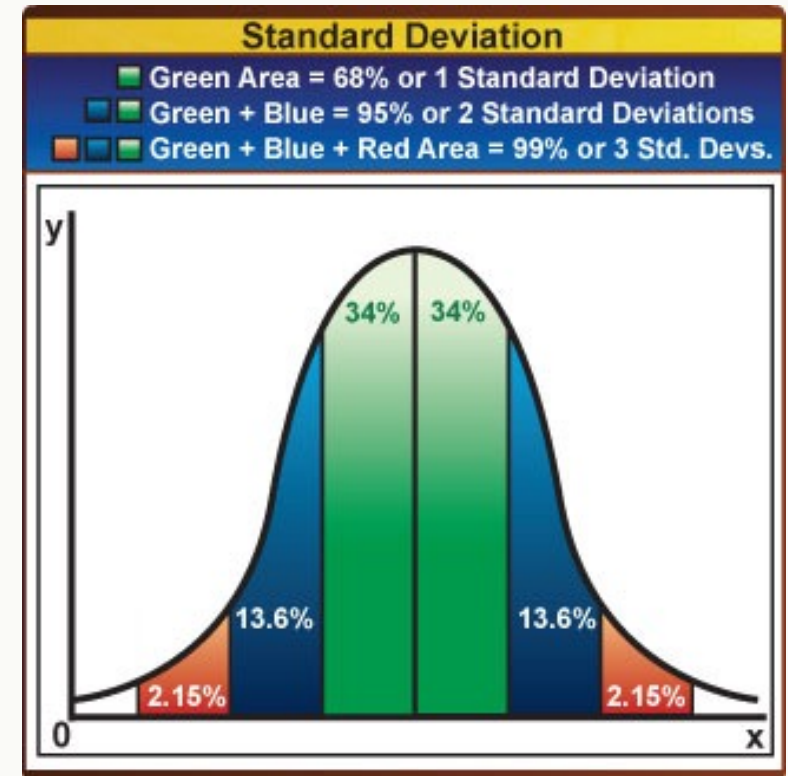
$$\bar{d} = \frac{1}{N-1} \sum_i |x_i - \bar{x}|$$

Estimating How Good the Data Are

Variance: (true*) Standard deviation, root mean square deviation (rms or sigma) which is the better estimator for random errors than the mean deviation

$$\sigma_x^2 = \frac{1}{N-1} \sum_i (x_i - \bar{x})^2$$
$$\sigma_x = \sqrt{\frac{1}{N-1} \sum_i (x_i - \bar{x})^2}$$

* This is called the “true” because we are using the values to determine the mean. Thus, we lose one “degree of freedom” when we calculate the mean from the data and hence the N-1. If the mean is known a priori, then N-1 -> N



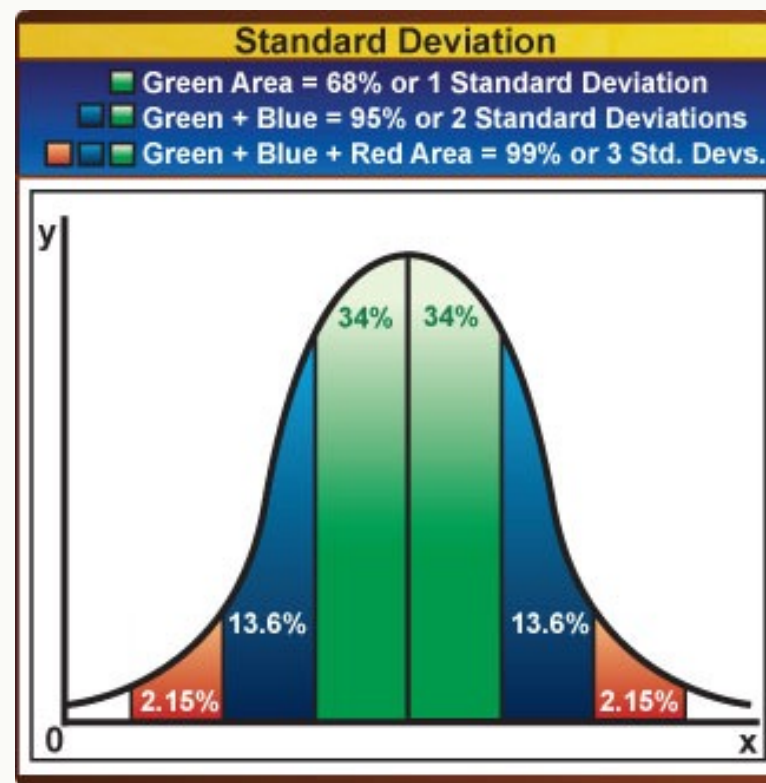
Estimating how Good your Data Are Cont.

You can use the equation of the normal distribution to determine the fraction of data points within 1σ , 2σ , 3σ , etc.

- 68% of points are within 1σ
- 95% of points are within 2σ
- 99.7% of points are within 3σ (3/1000 are not)
- 99.994% ... 4σ

Works well if your distribution is Gaussian, fails horribly if it is not.

These are sometimes called “confidence intervals”. E.g., with 95% confidence we can say the mean lies within 2-sigma of this value. But all distributions and not just Gaussians have confidence intervals.



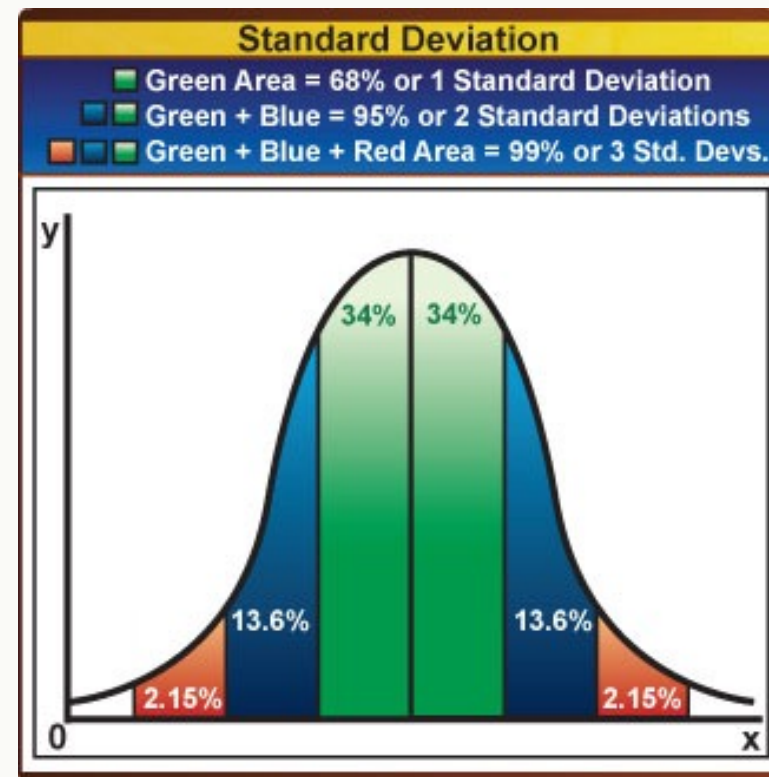
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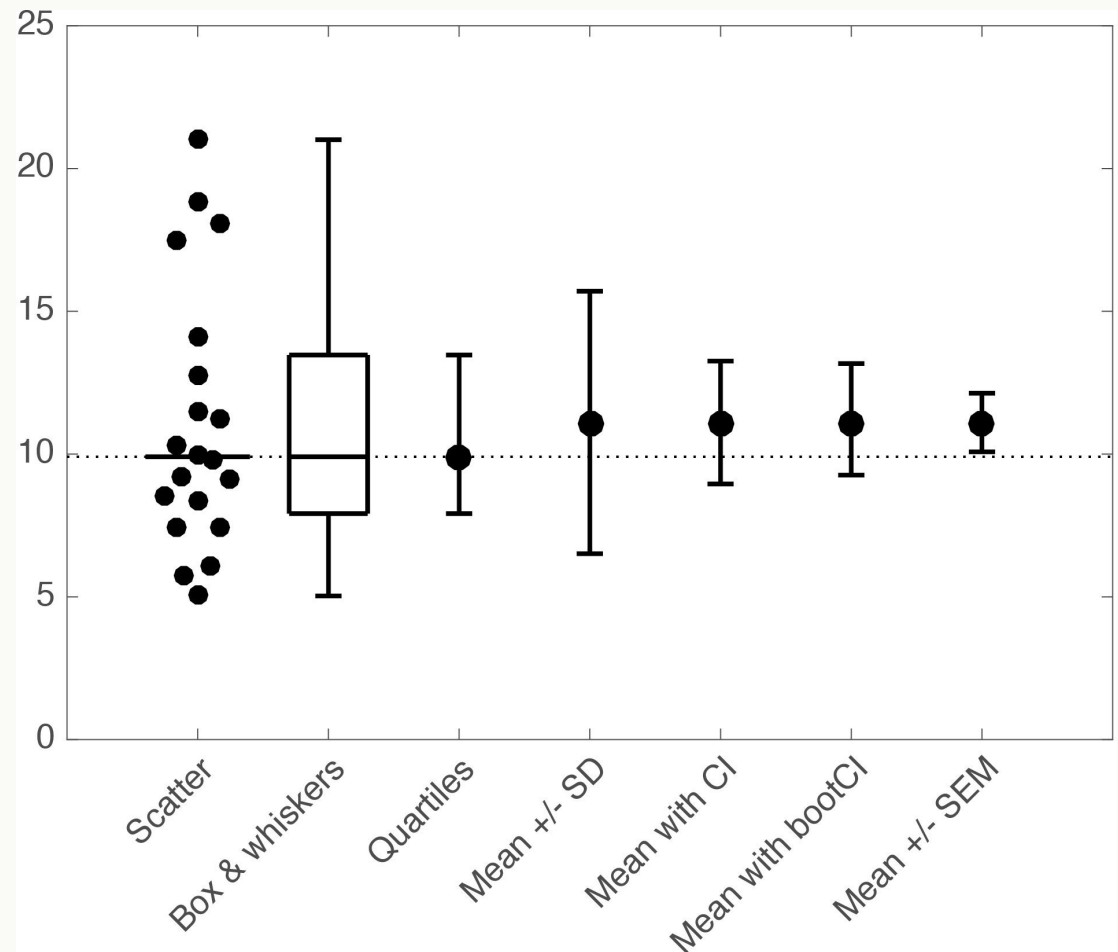
Works well if your distribution is Gaussian, fails horribly if it is not.

Example: when the "highly improbable" events are more probable than you estimated (stock market crash). Sometimes called the "long tail". Student t-distribution does a better job of capturing outliers than a Gaussian.



Estimating how Good your Data Are Cont.

There are many different ways to summarize and visualize your measurements:



Error Propagation

- Let's say you've measured multiple quantities and now you want to combine them in some mathematical formula.
- For example, I've measured the time of one transit of an exoplanet to be $t_1 \pm \sigma_1$ and I've measured the next transit to occur at $t_2 \pm \sigma_2$.
- What can I say the period of the orbiting exoplanet to be?

Error Propagation

- Let's say you've measured multiple quantities and now you want to combine them in some mathematical formula.
- For example, I've measured the time of one transit of an exoplanet to be $t_1 \pm \sigma_1$ and I've measured the next transit to occur at $t_2 \pm \sigma_2$.
- $P = t_2 - t_1$. But what about σ_p ?

Error Propagation

As another example. What if I've measured Planck's constant to be:

$$6.626070040 \pm 0.000000081 \times 10^{-34} \text{ J-s}$$

And the frequency of a photon to be $10.000 \pm 0.001 \text{ GHz}$

What is the energy of the photon ($E=hf$)?

What about the value for σ_E ?

Error Propagation – start with a simple linear function

$$f = \sum_i^n a_i x_i : f = ax$$

$f = P$

$x_1 = t_2$

$a_1 = 1$

$x_2 = t_1$

$a_2 = -1$

Given individual uncertainties on measurements x_i , which we denote σ_i , what is the total uncertainty on f ?

Error Propagation:

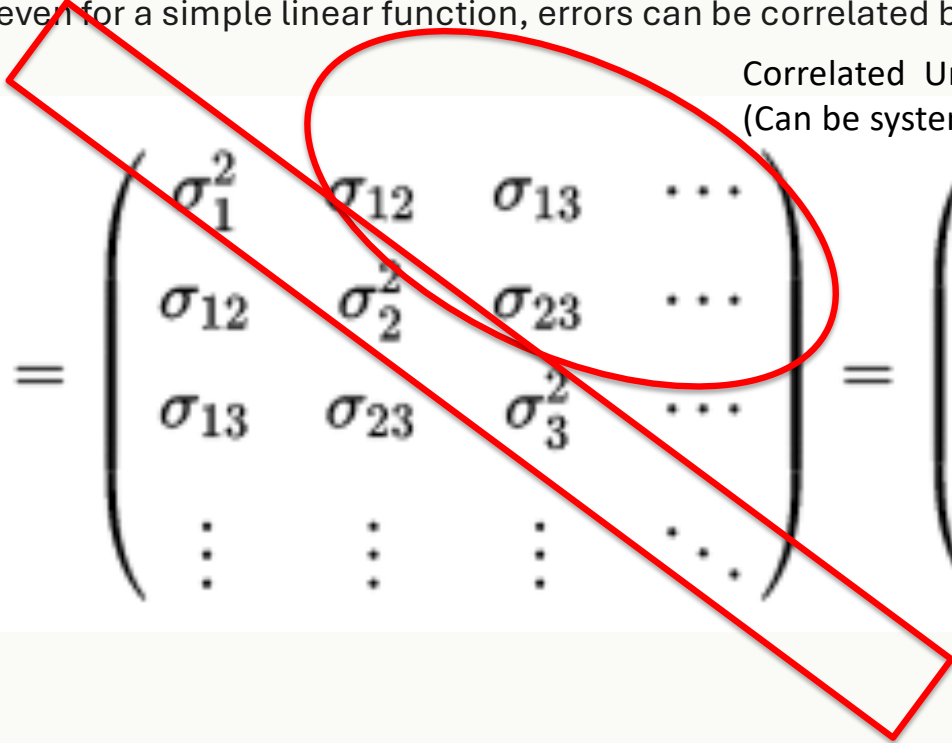
In reality, even for a simple linear function, errors can be correlated between variables:

$$\Sigma^x = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \cdots \\ \sigma_{12} & \sigma_2^2 & \sigma_{23} & \cdots \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} = \begin{pmatrix} \Sigma_1^x & \Sigma_{12}^x & \Sigma_{13}^x & \cdots \\ \Sigma_{12}^x & \Sigma_2^x & \Sigma_{23}^x & \cdots \\ \Sigma_{13}^x & \Sigma_{23}^x & \Sigma_3^x & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Error Propagation:

In reality, even for a simple linear function, errors can be correlated between variables:

Correlated Uncertainties
(Can be systematic or random)


$$\Sigma^x = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \cdots \\ \sigma_{12} & \sigma_2^2 & \sigma_{23} & \cdots \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} = \begin{pmatrix} \Sigma_1^x & \Sigma_{12}^x & \Sigma_{13}^x & \cdots \\ \Sigma_{12}^x & \Sigma_2^x & \Sigma_{23}^x & \cdots \\ \Sigma_{13}^x & \Sigma_{23}^x & \Sigma_3^x & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Error Propagation:

In reality, even for a simple linear function, errors can be correlated between variables:

$$\Sigma^x = \begin{pmatrix} \sigma_1^2 & 0 & \sigma_{13} & \cdots \\ 0 & \sigma_2^2 & \sigma_{23} & \cdots \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} = \begin{pmatrix} \Sigma_1^x & \Sigma_{12}^x & \Sigma_{13}^x & \cdots \\ \Sigma_{12}^x & \Sigma_2^x & \Sigma_{23}^x & \cdots \\ \Sigma_{13}^x & \Sigma_{23}^x & \Sigma_3^x & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Error Propagation:

$$f = \sum_i^n a_i x_i : f = \mathbf{a} \mathbf{x}$$

$$\sigma_f^2 = \sum_i^n \sum_j^n a_i \Sigma_{ij}^x a_j = \mathbf{a} \Sigma^x \mathbf{a}^\top$$

Error Propagation:

In the case that the variables in x are uncorrelated this simplifies further to

$$\sigma_f^2 = \sum_i^n a_i^2 \sigma_i^2.$$

In the simplest case of identical coefficients and variances, we find

$$\sigma_f = \sqrt{n} a \sigma.$$

Key result: increase number of measurements
and decrease uncertainty from random errors by $\sqrt{n}$:

For an average, $a_i = 1/n$ and thus: $\sigma_f = \sigma/\sqrt{n}$

Full Error Propagation Formula for Non-Linear Functions – linearize them as an approximation!

$$f_k \approx f_k^0 + \sum_i^n \frac{\partial f_k}{\partial x_i} x_i$$

$$\sigma_f^2 \approx \left| \frac{\partial f}{\partial a} \right|^2 \sigma_a^2 + \left| \frac{\partial f}{\partial b} \right|^2 \sigma_b^2 + 2 \frac{\partial f}{\partial a} \frac{\partial f}{\partial b} \sigma_{ab}.$$

Propagation of Errors

In these equations a and b are constants, u and v are variables

$$x = au \pm bv \quad \sigma_x^2 = a^2 \sigma_u^2 + b^2 \sigma_v^2 \pm ab \sigma_{uv}^2$$

$$x = \pm auv \quad \frac{\sigma_x^2}{x^2} = \frac{\sigma_u^2}{u^2} + \frac{\sigma_v^2}{v^2} + 2 \frac{\sigma_{uv}^2}{uv}$$

$$x = \pm \frac{au}{v} \quad \frac{\sigma_x^2}{x^2} = \frac{\sigma_u^2}{u^2} + \frac{\sigma_v^2}{v^2} - 2 \frac{\sigma_{uv}^2}{uv}$$

Propagation of Errors

In these equations a and b are constants, u and v are variables

$$x = au^{\pm b} \qquad \frac{\sigma_x}{x} = \pm b \frac{\sigma_u}{u}$$

$$x = ae^{\pm bu} \qquad \frac{\sigma_x}{x} = \pm b \sigma_u$$

$$x = a^{\pm bu} \qquad \frac{\sigma_x}{x} = \pm (b \ln a) \sigma_u$$

$$x = a \ln(\pm bu) \qquad \sigma_x = a \frac{\sigma_u}{u}$$

What does this mean?

The area of a rectangle is given by:

$$A = LW$$

You have the following measurements:

$$L = 22.1 \pm 0.1 \text{ cm} \quad W = 7.3 \pm 0.2 \text{ cm}$$

What is the uncertainty in A?

$$x = \pm auv \quad \frac{\sigma_x^2}{x^2} = \frac{\sigma_u^2}{u^2} + \frac{\sigma_v^2}{v^2} + 2 \frac{\sigma_{uv}}{uv}$$

$$\frac{\sigma_A^2}{A^2} = \frac{\sigma_L^2}{L^2} + \frac{\sigma_W^2}{W^2} = \frac{(0.1)^2}{(22.1)^2} + \frac{(0.2)^2}{(7.3)^2} = LW$$

$$\sigma_A = \sqrt{(22.1 \times 7.3)^2 [LW]}$$

Signal-to-Noise Ratio

Now that we understand errors/uncertainties, we can discuss a measure of the significance of an observational result.

Signal is the number of “detections.” For our purposes, at least to begin, signal will generally be the number of photons detected from our source.

Noise in this first case is “everything else” including photons from the “sky,” signal from the electronics, and random fluctuations due to photon detection being a poisson process.

In the simple case where there is no electronic or sky noise:

$$\frac{S}{N} = \frac{N_{photons}}{\sqrt{N_{photons}}}$$

Signal-to-Noise Cont.

Signal	Noise	S/N
10	3.16	3.16
100 (10^2)	10.00	10.00
1000 (10^3)	31.62	31.62
10000 (10^4)	100.00	100.00
100000 (10^5)	316.23	316.23

Note: if signal increases linearly with time observing time must be increased by a factor of 100 for S/N to increase by a factor of 10. This is what we mean when we say "The SNR goes as the square root of the integration time."